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Body capturing using adaptive cartesian grids

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1 Introduction

In recent years there has been considerable interest in using adaptive cartesian grids to compute complex flows (e.g. Berger & Collela 1989; Berger & Leveque 1990; Quirk, J.J. 1990). Such methods are extremely effective for the interior part of the flow, but they suffer from the disadvantage that they do not represent solid surfaces as accurately as body fitted grids. This difficulty can in part be overcome by using a polygonal representation of the body surface (Leveque 1988, Quirk 1992), but it is then impossible to avoid the presence of extremely small cells which lead to very small time steps when using explicit schemes. For inviscid problems this time step restriction can be avoided by allowing waves to pass through more than one cell in a time step (Leveque 1988, Berger & Leveque 1990), merging the small cells with neighbouring large ones (Quirk 1992) or using sub time stepping in the small cells. However, it is not easy to extend these ideas to viscous problems or to more than two dimensions, and it is therefore worth exploring alternative approaches.

It is much simpler to dispense with a polygonal representation of the body surface and to just use a series of steps. The problem with this is that in inviscid flows the steps generate O(1) errors which then propagate out into the flow. However, in viscous flow it is possible to confine these errors to the boundary provided the mesh Reynolds' number at the solid surface is sufficiently small. In fact it turns out that one can get extremely good results for a boundary layer flow provided there are about 5-10 cells in the boundary layer.

2 Adaptive cartesian grid

We use a hierarchical series of rectangular cartesian grids $G^0 \ldots G^N$, so that grid G^n has mesh spacing $\frac{h}{2^n}$. Grids G^0 and G^1 cover the entire

computational domain, but the others need only occupy part of it. Using a pseudo language, the basic integration is performed by

INTEGRATE $(n, t, \Delta t^0)$

where the control algorithm is essentially recursive:

procedure *INTEGRATE* $(n, t, \Delta t)$ **begin**
  MARCH $(n, t, \Delta t)$
  if $(n < N)$ **then**
    TEST(n)
    if (G^{n+1}) *exists* **then**
      INTEGRATE $(n+1, t, \frac{\Delta t}{2})$
      INTEGRATE $(n+1, t+\frac{\Delta t}{2}, \frac{\Delta t}{2})$
      RESTRICT(n)
    endif
  endif
end

MARCH advances the solution on grid G^n from time t to $t+\Delta t$; *TEST(n)* uses the solution on G^{n-1} (if $n > 0$) and G^n to decide whether cells on the latter need be refined. Unlike other authors, we do this on a cell by cell basis i.e. we do not attempt to collect fine cells into rectangular patches. *RESTRICT(n)* is the restriction operator of G^{n+1} to G^n . Of course, we unroll the recursion, so that the algorithm can be implemented, portably in Fortran 77.

Storing and integrating the solution on all grids not only gives us good error estimators, but also provides boundary conditions at the edges of incomplete fine grids.

3 Boundary layer on a flat plate

Given that even calculations with body fitted grids need about 10-20 cells to resolve the boundary layer, it is interesting to see what happens when we use a Cartesian grid at this resolution. Figure 1 shows an adaptive grid used for calculating the Blasius boundary layer on a flat plate inclined at an angle of 30° to the grid. Since we want to avoid the complications associated with the Navier-Stokes region at the leading edge, we start at a finite distance from the leading edge and input the Blasius solution at the left boundary. The parameters of the problem are :

Free stream	$p = 1$, $\rho = 1$, $u = 0.3$, (Mach No.$= 0.253$), $\gamma = 1.4$, kinematic viscosity $\nu = 2.4 \times 10^{-4}$.
Plate leading edge	$x = 0.0$, $y = 0.0$
Plate angle relative to grid	30°
Computational domain	$1.0 < x < 1.5$, $0.5 < y < 2.0$

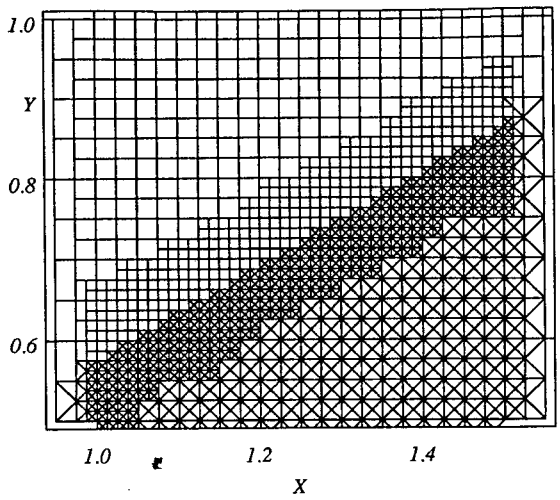


FIG. 1. Adaptive cartesian grid for flow past a flat plate. The hatched area is the solid and the flow is from left to right.

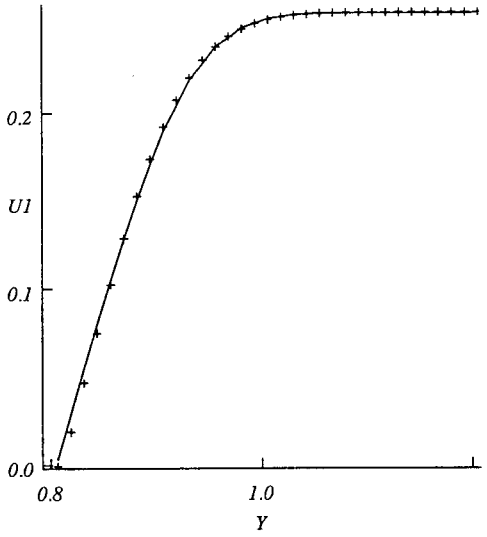


FIG. 2. x-velocity at $x = 1.4$ for flow past a flate plate at an angle of 30° to the grid. – Blasius solution, × numerical solution.

Figure 2 shows a comparison between the computed results and the Blasius solution at $x = 1.4$. It can be seen that the agreement is surprisingly good given that there are only 10 cells in the boundary layer. Although there are O(1) errors in the velocity near the body surface, the error in the velocity

gradient is extremely small, except for the two cells next to the wall. This means that the error in the skin friction is also small.

In order to get these results, we have had to modify the boundary conditions at the wall. Instead of solving a Rieman problem with zero velocity at the wall, we set the pressure at the wall equal to that in the cell next to the wall. One might think that this would lead to large errors for shocks colliding with rigid walls, but in fact it only increases the entropy error by about 20%.

4 Steady supersonic flow past a wedge

The previous calculation shows that we can accurately compute an essentially incompressible boundary layer without requiring excessive resolution. There are, however, cases where we are not interested in the structure of the boundary layer and for these the viscosity only needs to be large enough to prevent the errors at the wall from polluting the rest of the solution.

Figures 3 and 4 show the results for steady supersonic flow past a wedge. The parameters of the problem are :

Free stream	$p = 1$, $\rho = 1$, $u = 5$, (Mach No.$= 4.23$),
	$\gamma = 1.4$, kinematic viscosity $\nu = 1.25 \times 10^{-3}$.
Wedge leading edge	$x = 0$, $y = 0$
Wedge angle	15°
Computational domain	$-1 < x < 5$, $0 < y < 4$

Apart from a high pressure region near the leading edge, the computed solution agrees very well with the inviscid one. The shock angle is 27° , compared with the theoretical one of 26.3^o. The thickness of the boundary layer at $x = 4$ is 0.1 and one can see from Figure 4 that the streamlines are very nearly parallel to the wedge even close to the wall.

5 Mach 2.81 shock incident on a cylinder

This problem has been computed by Yang, Yen and Lomax (1987) using cylindrical polar coordinates, by Leveque (1988) and Quirk (1992) using a Cartesian grid with partial cells. The results can be compared with the Schlieren photographs in Bryson and Gross (1961). The parameters are :

Preshock	$p = 1$, $\rho = 1$, $u = 0.0$,
	$\gamma = 1.4$, kinematic viscosity $\nu = 4 \times 10^{-3}$.
Postshock	$p = 9.11$, $\rho = 3.68$, $u = 2.43$,
Computational domain	$-4 < x < 6$, $0 < y < 6$

Figures 5 and 6 show the grid and pressure contours. These should be compared with figure 3 in Bryson and Gross. The shapes of the shocks agree

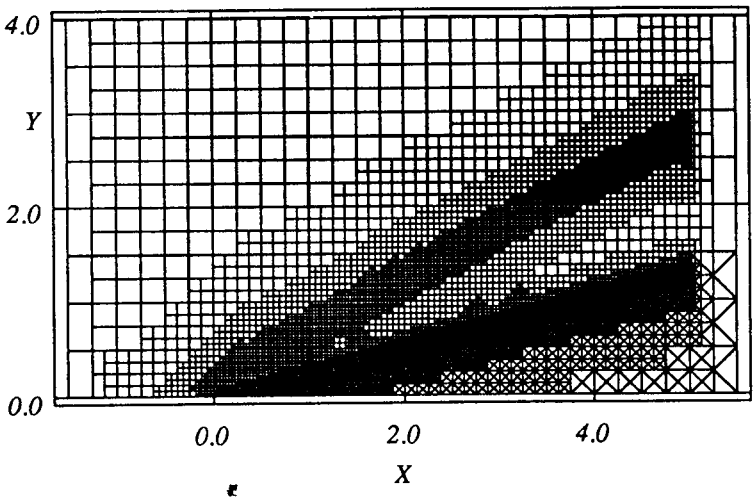


FIG. 3. Grid for steady supersonic flow past a 15° wedge.

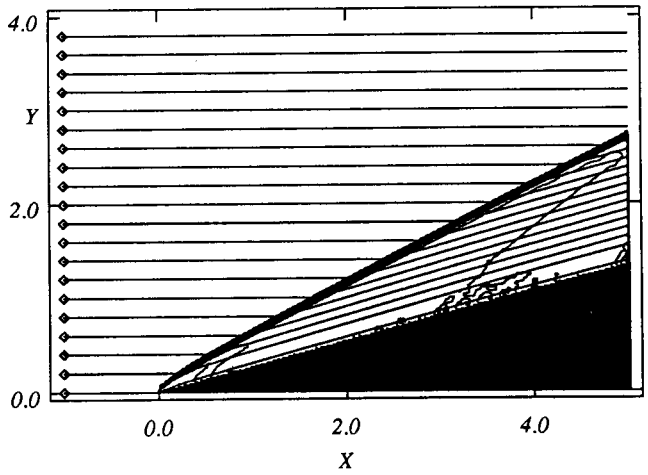


FIG. 4. Pressure contours and streamlines for steady supersonic flow past a 15° wedge.

well with the Schlieren photographs and the numerical noise is confined to a very small region near the cylinder surface.

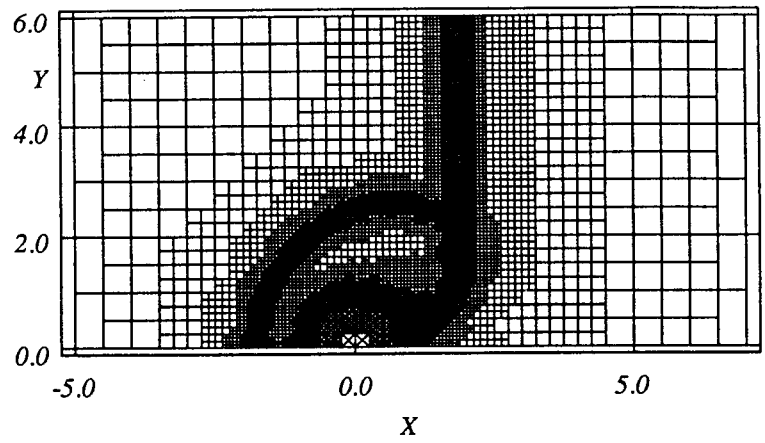


FIG. 5. Grid for a Mach 2.82 shock incident on a cylinder.

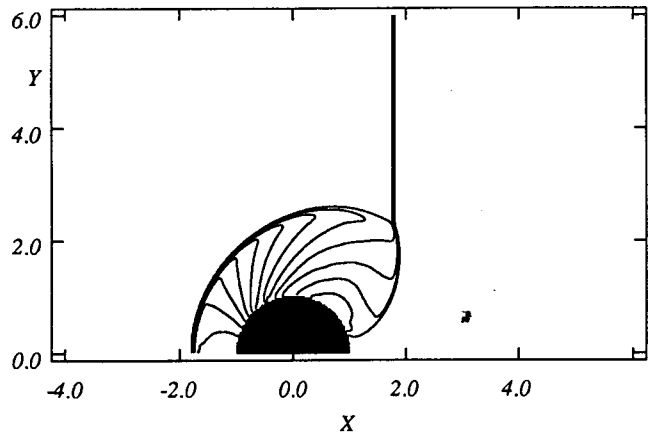


FIG. 6. Pressure contours for a Mach 2.82 shock incident on a cylinder.

6 Conclusions

These results suggest that a Cartesian grid is more efficient than a body fitted one for many problems. Quite apart from the saving in grid generation costs, numerical algorithms are much simpler on Cartesian grids and one avoids the difficulties associated with the highly distorted cells that occur when grids are fitted to solids with complicated shapes. With a Cartesian grid it is also easy to implement operator split diagonalised implicit

schemes (Marx & Piquet 1988). The results in Figure 2 were obtained with such a scheme.

In a sense this can be regarded as body capturing in that, like shock capturing, we tolerate O(1) errors at the solid surface but use an algorithm that ensures that these errors do not pollute the rest of the solution.

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